1) $f(x) = x^2 - 6x + 5$

$f'(x) = 2x - 6$
$f''(x) = 2$

$x_0 = 0$

$x_1 = 0 - \frac{f'(0)}{f''(0)} = -\left(\frac{-6}{2}\right) = \boxed{3}$

minima

2) $f(x) = x^3 - 3x + 1$

$f'(x) = 3x^2 - 3$
$f''(x) = 6x$

$x_0 = 2$
$x_1 = 2 - f'(2)/f''(2) = 2 - 9/12 = 1.25$
$x_2 = \frac{5}{4} - \frac{f'(5/4)}{f''(5/4)} = 1.025$

rate ↓

3) $f(x) = ax^2 + bx + c$

$f'(x) = 2ax + b$

$f''(x) = 2a$

minina = $-b/2a$

x_0

$$x_1 = x_0 - \frac{2ax_0 + b}{2a}$$

$\boxed{x_1 = -b/2a}$

so irrespective of x_0 $x_1 = -b/2a$

which is minima.

(4) $f(x,y) = 2x^2 + 4y^2 - 3x + y$

$f'_x(x,y) = 4x - 3$

$f'_y(x,y) = 8y + 1$

$f''_{xx}(x,y) = 4$

$f''_{yy}(x,y) = 8$

$f''_{xy}(x,y) = 0$

$H = \begin{bmatrix} 4 & 0 \\ 0 & 8 \end{bmatrix}$ positive definite.

$\lambda = 4, 8$

5

$$H = \begin{bmatrix} 4 & 0 \\ 0 & 8 \end{bmatrix}$$

$$H^{-1} = \begin{bmatrix} 1/4 & 0 \\ 0 & 1/8 \end{bmatrix}$$

$$x, y = (0, 0)$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/4 & 0 \\ 0 & 1/8 \end{bmatrix} \nabla f$$

$$\nabla f_{(0,0)} = \begin{bmatrix} 4x-3 \\ 8y+1 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/4 & 0 \\ 0 & 1/8 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

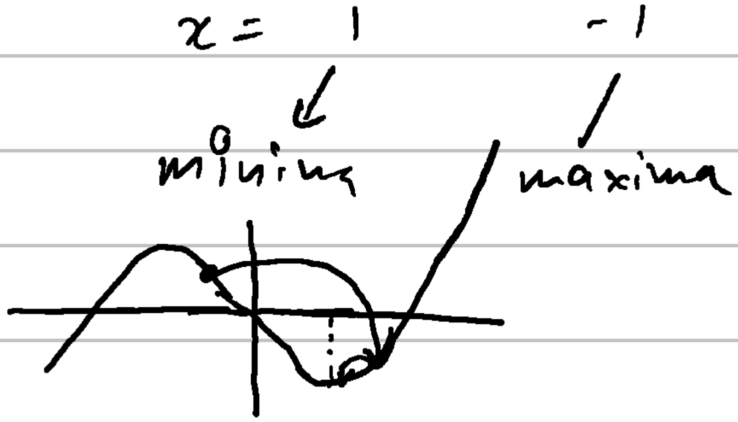
$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = - \begin{bmatrix} -3/4 \\ 1/8 \end{bmatrix} = \begin{bmatrix} 3/4 \\ -1/8 \end{bmatrix}$$

Calc mistake (Basin of attraction issue)
↳ Read about it

(6) $f(x) = x^3 - 3x$

$f'(x) = 3x^2 - 3$

$f''(x) = 6x$



$x_0 = -0.5$

$f'(0.5) = 3 \cdot \frac{1}{4} - 3 = \frac{3}{4} - 3 = -\frac{9}{4}$

$f''(0.5) = 6 \cdot \frac{1}{2} = 3$ → This should be -3 & x_1 moves toward maxima

$x_1 = x_0 - \left(\frac{-\cancel{9}^{3}/4}{\cancel{3}} \right)$

$x_1 = \frac{2}{4} + \frac{3}{4} = \frac{5}{4}$

- x_1 overshoots the minimum value

$x_2 = 1.025$ → closer to minima.

7 $$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

$$Q(x) = f(x_n) + f'(x_n)(x - x_n) + \frac{1}{2} f''(x_n)(x - x_n)^2$$

$$Q'(x) = f'(x_n) + f''(x)(x - x_n) = 0$$

$$x - x_n = -\frac{f'(x_n)}{f''(x_n)}$$

$$\boxed{x = x_n - \frac{f'(x)}{f''(x_n}}$$

Same as newton's method.

8) $\dfrac{1}{f''(x)}$ can be seen as an adaptive learning rate.

⇒ Newton's method would be preffered where we need to reach the minima in less number of steps

⇒ • Gradient descent would be preffered when $f''(x)$ is computationally very expensive to compute.

• Where there are a lot of variables & H^{-1} is expensive to compute.

(9) $f(x) = x^p$ $\qquad p \geq 2$ $\to$ even integer.

$x_0 \neq 0$

$$f'(x) = p\, x^{p-1}$$
$$f''(x) = p \cdot (p-1)\, x^{p-2}$$

a) $$x_{n+1} =: x_n - \frac{\cancel{p} \cdot x_n^{\cancel{p-1}}}{\cancel{p} \cdot (p-1)\, x_n^{\cancel{p-2}}}$$

$$= x_n - \frac{x_n}{p-1}$$

$$= \left(\frac{p-1-1}{p-1}\right) x_n$$

$$= \left(\frac{p-2}{p-1}\right) x_n$$

b) $x_{n+1} = \left(\frac{p-2}{p-1}\right) x_n$

$$x_{n+1} = \left(\frac{p-2}{p-1}\right)^n x_0$$

c) For x_{n+1} to converge in a single step and reach minima i.e. $x = 0$

$$\frac{p-2}{p-1} = 0$$

$$\boxed{p = 2}$$

d) For $p = 4$

$$x_n = \left(\frac{4-2}{4-1}\right)^n x_0$$

$$x_n = \left(\frac{2}{3}\right)^n x_0$$

$\Rightarrow$ This sequence is converging linearly with a rate $2/3$.

Newton's method's standard theorem guarantees quadratic convergence under 3 conditions.

1. Starting point sufficiently close ✓
2. Continuous & twice differentiable ✓
3. $f'(x) \neq 0$ at root ✗

If $f''(x) = 0$ at root then the rate of convergence drops to linear.

Had to read about quadratic / linear convergence.

(10) $f(x) = x^3 - 3x$

$f'(x) = 3x^2 - 3$
$f''(x) = 6x$

a) $N(x) = x - \frac{3x^2 - 3}{6x}$

$= \frac{6x^2 - 3x^2 + 3}{6x} = \frac{3x^2 + 3}{6x}$

$= \frac{x^2 + 1}{2x}$

b) $N(x^*) = x^*$

$\frac{x^2 + 1}{2x} = x$

$2x^2 = x^2 + 1$

$x^2 - 1 = 0$ $\boxed{x = \pm 1}$

They are the critical points.

$\Rightarrow$ If $N(x) = x$, this means that the next step in newtons method would yield the same numbers.

$\downarrow$

We have reached our points of interest.

c) $N(x) = \dfrac{x^2+1}{2x}$

$$N'(x) = \frac{2x}{2x} - \frac{1}{2} \cdot \frac{x^2+1}{x^2}$$

$$N'(x) = 1 - \frac{x^2+1}{2x^2}$$

$$N'(x) = 1 - \frac{x^2+1}{2x^2}$$

$$N'(1) = 1 - \frac{2}{2} = 0$$

$$N'(-1) = 0$$

$$e_{k+1} \approx \underbrace{N'(x^*)}_{\downarrow \; 0} e_k + \frac{N''(x^*)}{2} e_k^2$$

$$e_{k+1} \approx \frac{N''(x^*)}{2} e_k^2$$

quadratic convergence.

$$N''(x) = 1 - \frac{1}{2} - \frac{1}{2x^2} = \frac{1}{x^3}$$

$$N''(1) = 1$$

$$e_{k+1} \approx \frac{N''(1)}{2} e_k^2$$

$$\approx \left(\frac{1}{2}\right) e_k^2$$

or near -1

$$\left(-\frac{1}{2}\right) e_k^2$$

d) $f''(x) = 6x$

$f''(x) = 0$ at $x = 0$

So, at $x = 0$ newton's step is undefined.

$$Q(x) = \underline{f(0)} + \underline{f'(0)(x-0)} + \frac{1}{2} \underbrace{f''(0)}_{0} (x-0)^2 \ldots$$

all zeros

$$Q_x = 0 + (-3x) = \underline{\underline{-3x}}$$

Hence newton's method is not defined at inflection points.